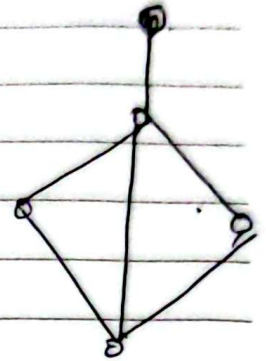
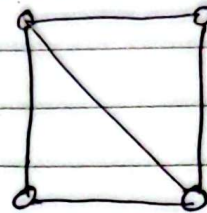
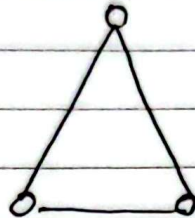


Notes

Concept of Graph Colouring

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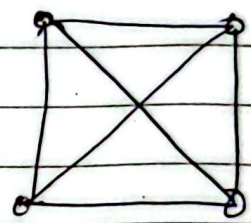
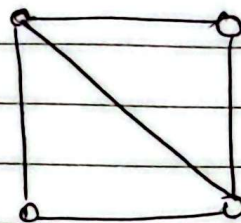
Painting all the vertices of a graph with colours such that no two adjacent vertices have same colour is called colouring of graph.



Chromatic Number

The least number of colours reqd. for colouring of a graph G is called its chromatic number.

Example:



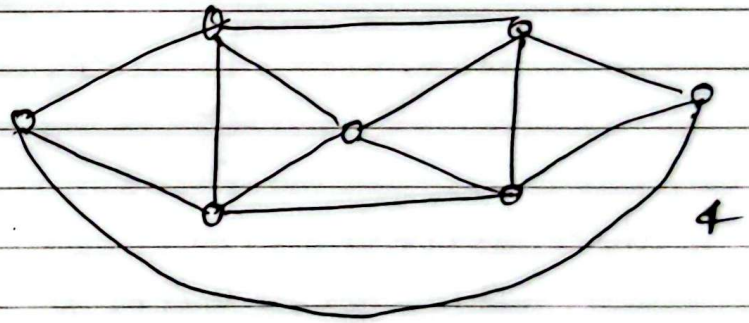
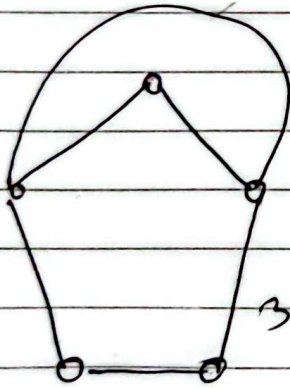
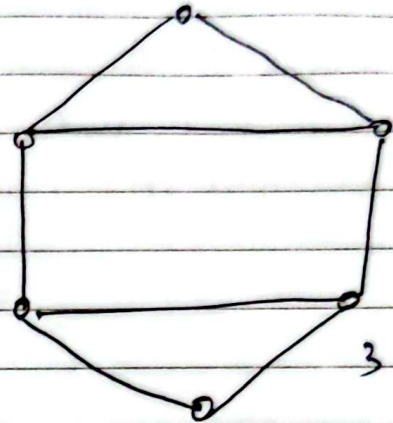
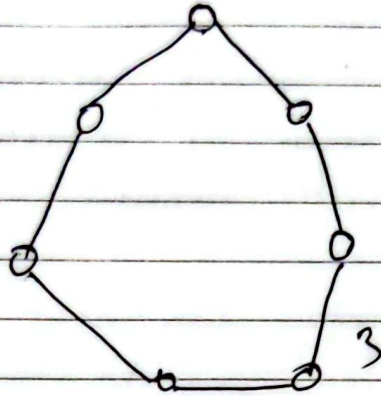
Note:

1. The chromatic number of graph G is denoted by $\chi(G)$.
2. If $\chi(G) = k$, then the graph is called k -chromatic.
3. Chromatic number of null graph is 1.
4. Chromatic number of complete graph K_n of n vertices is n .
5. If a graph is circuit with n vertices then
 - (i) It is 2-chromatic if n is even.
 - (ii) It is 3-chromatic if n is odd.

Notes

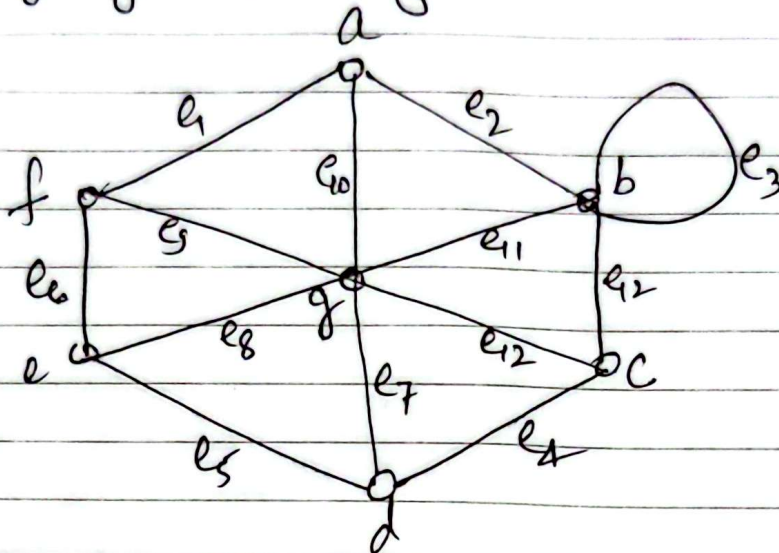
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Que. 1 Determine the chromatic numbers of graph.



Eulerian Graph and Hamiltonian Graph

1. Walk: A walk is a finite alternating sequence $v_1, e_1, v_2, e_2, v_3, e_3, \dots, e_n, v_n$ of vertices and edges, beginning and ending with same or different vertices.



then $a, e_1, f, e_9, g, e_{10}, b, e_3, e_{12}, c$ is walk.
 $a, e_{10}, g, e_9, f, e_1, a, e_2, b$ is walk.
 $a, e_2, b, e_{11}, g, e_{10}, a$ is walk.

(i) length of walk: The length of walk is the number of edges in it.

(ii) Closed and Open walk: A walk is said to be closed if its origin and terminous vertex ($v_0 = v_n$) is equal otherwise it is called open walk.
 e.g. $f, e_9, g, e_{10}, a, e_1, f$ is closed walk.

2. Trail: Any walk having different edges is called trail.

3. Circuit: A closed trail is called circuit.
 e.g. $a, e_1, f, e_9, g, e_{10}, a$

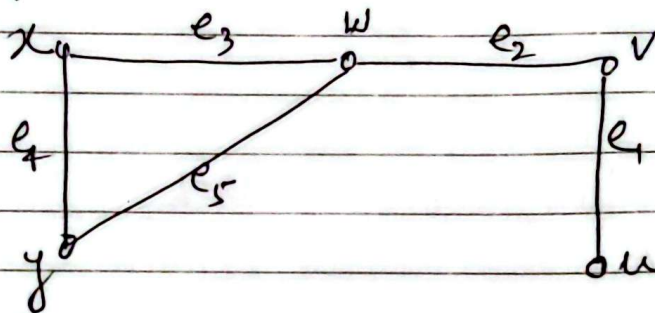
4. Path: A walk is called path if all vertices are not repeated.

5. Cycle: A closed path is called cycle.

1 (a) Eulerian Path

A path in a graph is said to be an Eulerian path if it traverses each edge in the graph once and only once.

e.g.

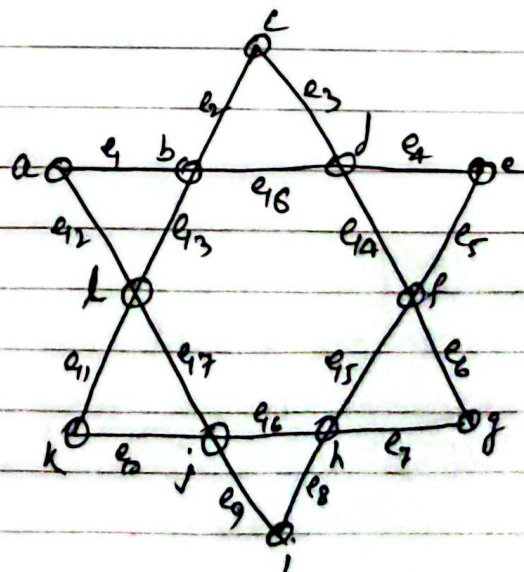


$u e_1 v e_2 w e_5 y e_4 x e_3 w$

(b) Eulerian Circuit

A circuit in a graph is said to be an Eulerian circuit if it traverses each edge in the graph once and only once.

e.g.

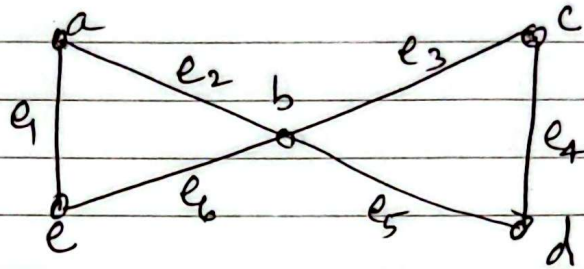


(c) Eulerian Graph

A connected graph which contains an Eulerian circuit is called Eulerian Graph.

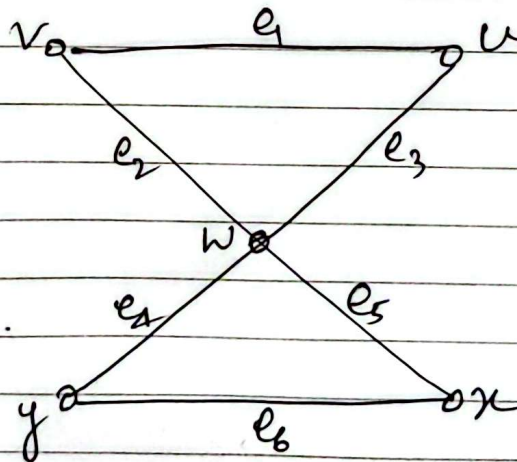
e.g. (i) previous figure

(ii)



2 (a) Hamiltonian Path

A path which contains every vertex of a graph G exactly once is called Hamiltonian graph.



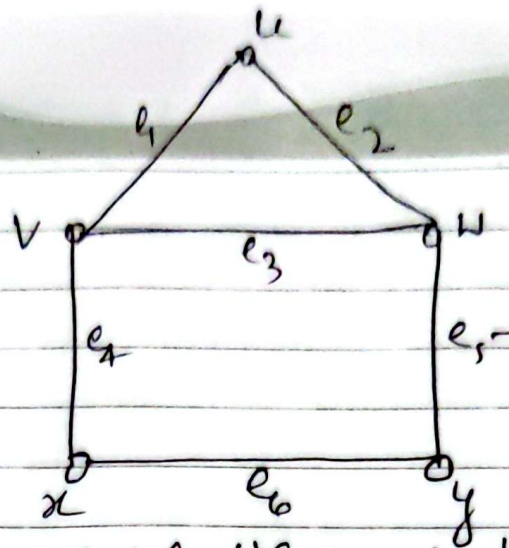
Here,

$u, v, e_2, w, e_5, x, e_6, y$
is Hamiltonian path.

(b) Hamiltonian Circuit

A circuit that passes through each of the vertices in a graph G exactly once except the starting vertex & end vertex is called Hamiltonian circuit.

Notes

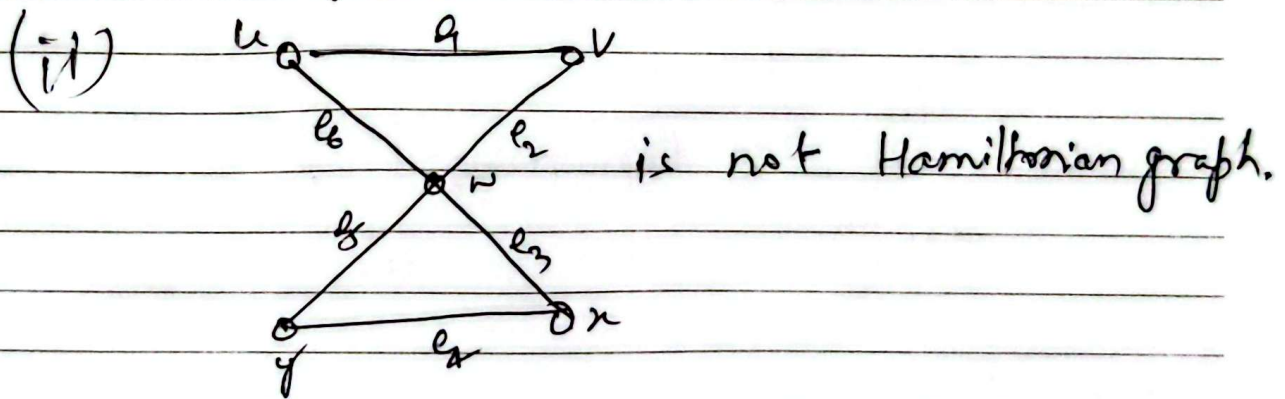


Here $ue_1ve_4xe_6ye_5we_2u$ is Hamiltonian circuit.
(~~2nd~~ edge को नजर में रखें)

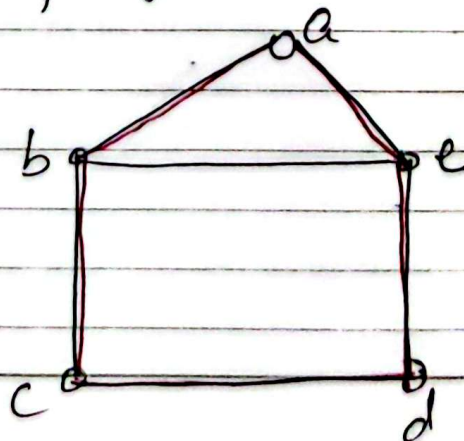
(C) Hamiltonian Graph:

A connected graph which contains Hamiltonian circuit is called Hamiltonian Graph.

e.g. (i) The above figure is an example of Hamiltonian graph.



Que 1. Determine a minimum Hamiltonian circuit for the graph given below.



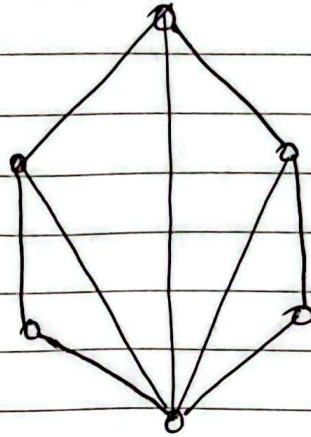
min. Hamit. circuit = 1.

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Que 2. Draw a graph with six vertices containing a Hamiltonian circuit but not Eulerian circuit.

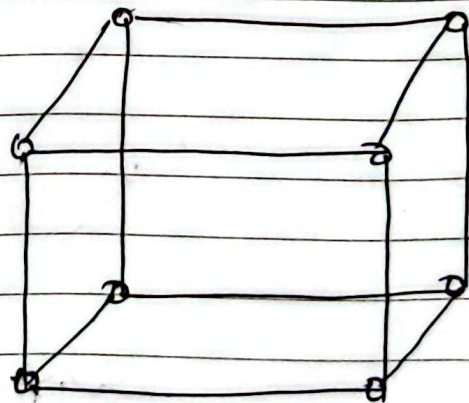
Sol: (यह Case में vertices को संयोजित edges को संख्या की कुल 5 है।)



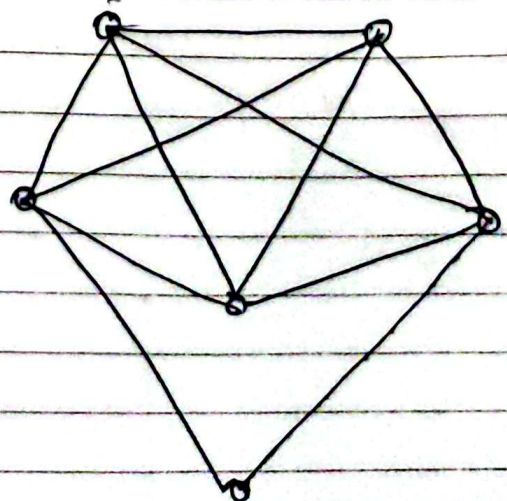
Que 3. Check the graph is Hamiltonian or Euler.

Sol: $\frac{\text{Sum no. of edges}}{\text{no. of vertices}}$

Hence, it is a Hamiltonian graph.



Que 4: The graph is both Euler as well as Hamiltonian.

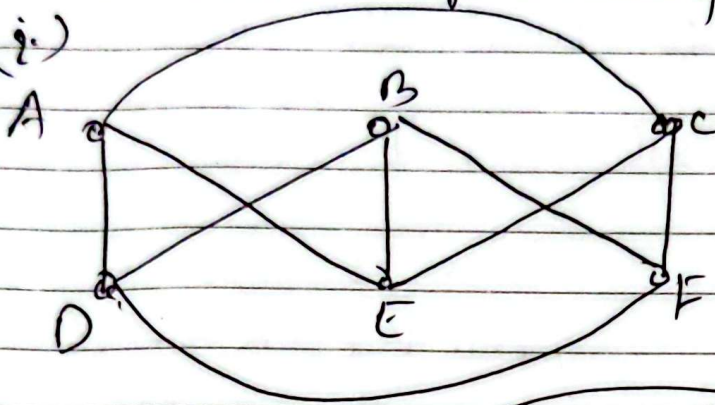


Notes

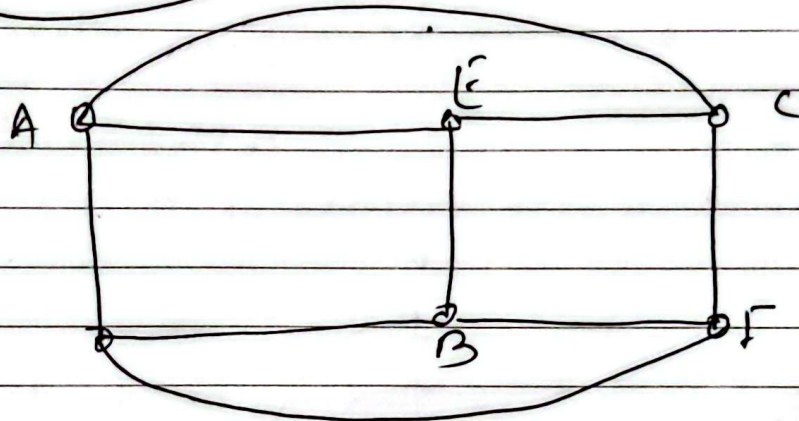
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Que 5: Draw a planar representation of each graph.

(i.)

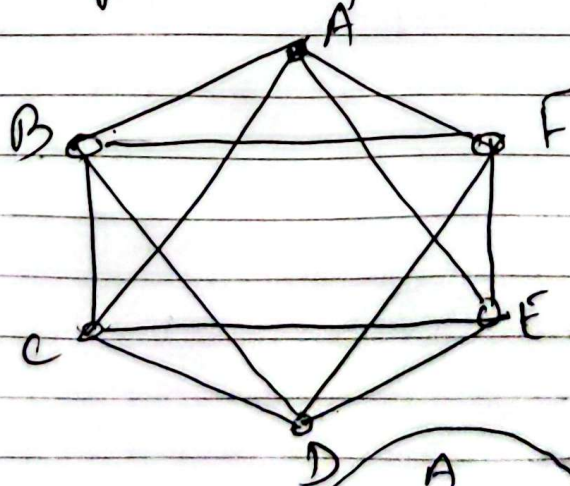


Solⁿ:

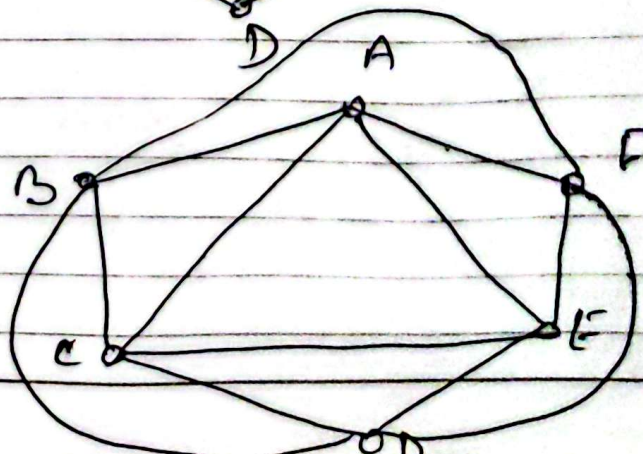


Redrawing the positions of the vertices B and E, we get a planar representation of the graph.

(ii)



Solⁿ:



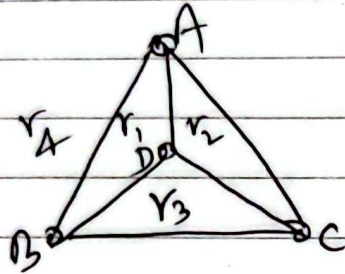
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Theorem: The sum of the degrees of the regions of a map is equal to twice the number of edges.

Theorem: (Euler's Formula): $V - E + R = 2$.

Ex.



Here, $V = 4, E = 6, R = 4$

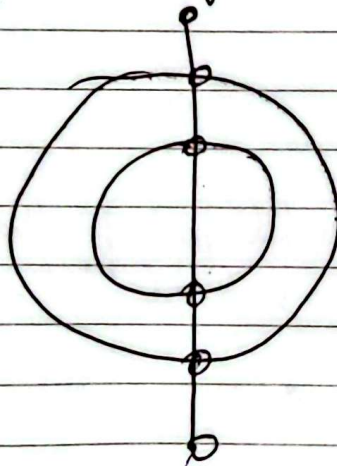
Hence $V - E + R = 4 - 6 + 4 = 2$

degree of regions $d(r_1) = 3$

$d(r_2) = 3, d(r_3) = 3, d(r_4) = 3$

$\therefore$ sum of the degrees of map $= 3 + 3 + 3 + 3 = 12$
 $= 2 \times \text{no. of edges.}$

Ex

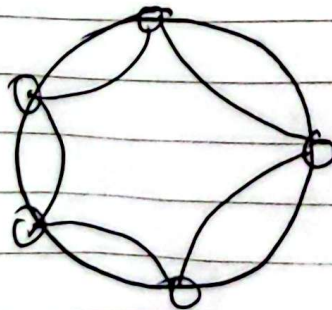


$V = 6, E = 9, R = 5$

$\therefore V - E + R = 6 - 9 + 5 = 2$

degree of outside region
 $= 6$ since two edges
are counted twice.

Ex



$V = 5, E = 10, R = 7$

$\therefore V - E + R = 2$

degree of outside region $= 5$.